SCHRÖDINGER EVOLUTION AS THE $\alpha \to 1^-$ COLLAPSE OF THE CAUSAL ACCUMULATION LAW: DERIVED UNITARITY AND THE LIGHT-CONE DISPERSION

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Abstract. We impose the Causal Accumulation Law (CAL) as a field equation, $T = 0$, with a power-law memory under causal Weyl integration and the channel orientation of the parent framework carried by the memory channel. As the fractional order approaches $\alpha = 1$ from below, the field equation collapses to the Schrödinger equation. The collapse takes place in a double scaling in which the memory weight $(1 - \alpha)$ vanishes while the operator it multiplies is held fixed.

Unitarity is derived, not posited. For every $\alpha < 1$, the bounded solutions on the whole time line form a strongly continuous unitary group, because the channel phase cancels the phase of the Weyl symbol on one sign of frequency. This replaces the unitarity posit of the companion paper.

The limit is one-sided for three reasons. The Weyl integral converges on oscillating carriers only for $\alpha < 1$. The memory-generated coefficients change sign across $\alpha = 1$. And from above, the response to a source either runs away at rate $\pi|1 - \alpha|\,|E|/\hbar$ or precedes its source, which is the Abraham–Lorentz–Dirac dichotomy.

The kernel-blind skeleton at $\alpha = 1$ and the Schrödinger edge are the two scalings of a single limit. That limit is Markovian in its evolution but not through a δ-kernel. At $\alpha = 1$ the kernel is the flat accumulator, whose inverse is ∂_t.

The past light cone still matters, because a causal δ-kernel has no spatial reach and so produces no kinetic term. For memory supported inside the cone, the $\alpha \to 1^-$ dispersion is exactly $E(k) = \hbar k v \coth(\hbar k v/E_0)$. It is causal for every k, and its slow-particle limit is Schrödinger with $m = 3E_0/(2v_{\text{eff}}^2)$. The zero-momentum energy generated by causal memory is therefore at most $\frac{2}{3}mc^2$.

Numerical runs of the field equation, using Weyl integration only, confirm each statement.

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ORCID: 0009-0000-0133-2291. Companion to *The Causal Accumulation Law: The Four Corners of α, Algebraic Closure, and the Standard Model* [1] and to *The $\alpha = 1$ Unitary Skeleton and the Emergence of Schrödinger Evolution in the δ-Kernel Limit of the CAL* [2], whose unitarity posit it replaces. Dated November 11, 2026.

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1. Scope and claim

The companion paper [2] obtains the Schrödinger form at the edge of the (α, kernel) table adjoining $\alpha = 1$ from below, but it consumes an input: Posit 1 assumes a unitary flow $U(t) = e^{i\theta(t)G}$ and then defines $\hat{H} \equiv i\hbar\dot{U}U^{-1}$. That step establishes the form of the equation only after unitarity and a generator have been assumed. The present paper removes the posit. The accumulation law itself, read as a field equation, collapses to the Schrödinger equation, and unitarity, the one-sidedness of the limit, and the mass formula follow.

> *Impose $T = 0$ with a power-law memory under causal Weyl integration, and let the memory channel carry the orientation $e^{-i\sigma\alpha\pi/2}$ of Axiom 3. For every $\alpha < 1$, the bounded solutions on the whole time line evolve by a unitary group (Theorem 5.1). As $\alpha \to 1^-$ with $(1-\alpha)\hat{R}_{\mathcal{H}}$ held fixed, they converge to Schrödinger evolution (Theorem 5.3). From above, the response to a source runs away or precedes the source (Theorem 5.7). A causal δ-kernel cannot generate a kinetic term (Lemma 6.1). For memory inside the past light cone, the limiting dispersion is $E = \hbar kv \coth(\hbar kv/E_0)$, causal for all k, with Schrödinger as its slow-particle limit (Theorem 6.4).*

All dynamics in this paper use Weyl integration over the whole past, which is the integral the accumulation law writes. Initial-value (Caputo) problems are not used. A Caputo problem is the Weyl derivative of a history frozen at its initial value for all earlier times. Such a frozen past has zero-frequency content, on which the Weyl integral of the field equation diverges, so a Caputo problem is not a solution of Axiom 3. Starting a solution is instead modelled by a source acting on a zero past (Proposition 5.5).

Bookkeeping. The inputs are:

- the field equation (Axiom 3), which replaces Posit 1 of [2];
- the kernel class (Definition 2.2);
- the double scaling (Definition 2.5).

Given these inputs, the following are derived and unconditional:

- Lemmas 4.1–4.4;
- Theorems 5.1, 5.3, 5.7 and 6.4;
- Propositions 5.5, 6.5 and 6.7, and Corollaries 5.4 and 6.6.

The following are not claimed:

- a value for the energy scale E_0, and so for the mass;
- Lorentz invariance of the light-cone dispersion, which is defined in the foliation frame;
- gauge coupling and position-dependent potentials beyond a constant offset;
- anything about measurement, which lives at $\alpha = 2$ [1, 3].

2. Definitions

Throughout, $\varepsilon \equiv 1 - \alpha$. Complex powers take the principal branch, and $E_c \equiv \hbar/\tau_c$.

Definition 2.1 (Causal Weyl integral and admissibility)**.** For a bounded $f : \mathbb{R} \to \mathcal{V}$ and $\alpha \in (0, 1)$, the causal (left-sided) Weyl integral is

$$W^{\alpha} f(t) = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} s^{\alpha-1} f(t-s)\, ds. \tag{1}$$

It integrates over the past only. The right-sided integral $\int_t^\infty$, which some texts also call Weyl's [4], integrates over the future and is excluded.

A function f is *admissible* if its time Fourier transform, as a tempered distribution, has no support at zero frequency. On admissible functions, D^α denotes the inverse of W^α. On a carrier $e^{-iEt/\hbar}$ the symbols are $W^\alpha \mapsto (-iE/\hbar)^{-\alpha}$ and $D^\alpha \mapsto (-iE/\hbar)^\alpha$.

Definition 2.2 (Memory kernel)**.** The memory kernel is

$$k(\tau, y) = \tilde{r}_1 \, \tau_c^{-\alpha} \, \frac{\tau^{\alpha-1}}{\Gamma(\alpha)} \, g_\tau(y),$$

dressed by the real transport $\mathcal{P}$. Here $\tau > 0$ is the lag, $\tilde{r}_1$ is the coincidence weight of the fiber factors, and $g_\tau \geq 0$ is a normalized isotropic spatial profile. Two classes are used.

(1) *Factorized:* $g_\tau = g$ does not depend on τ.
(2) *Light-cone supported:* $\operatorname{supp} g_\tau \subset \{|y| \leq c\tau\}$. Within this class:
   - a profile is *self-similar* if $g_\tau(y) = \tau^{-3} h(|y|/\tau)$, which makes it a superposition of thin shells of speed $u \in [0, c]$ with weight $w(u) \geq 0$, $\int w = 1$;
   - the *thin shell* is $w = \delta(u - v)$, with $v \leq c$.

   For a self-similar profile define $v_{\text{eff}}^2 \equiv \langle |y|^2 \rangle_{g_\tau} / \tau^2 = \int u^2 w(u)\, du \leq c^2$.

Definition 2.3 (Channel orientation)**.** The orientation is $\sigma \in \{+1, -1\}$. By Axiom 3 of [2], the memory channel carries the phase $e^{-i\sigma\theta}$ relative to the present channel, with $\theta = \alpha\pi/2$.

Definition 2.4 (Hilbert sector)**.** The spaces $\mathcal{V}$ and $\mathcal{H}$, the projection $\Pi_{\mathcal{H}}$ and the map f_{map} are those of [2, Defs. 2.3–2.5].

On the spinless, gauge-neutral sector $\mathcal{H} = L^2(\mathbb{R}^3)$, translation and rotation invariance give the spatial part of the factorized memory the form $\hat{R}_{\mathcal{H}} = r_0 \nabla^2 + r_1$ [2, Thm. 6.1(ii)]. For $\tilde{r}_1 > 0$ this has $r_1 = \tilde{r}_1$ and $r_0 = \tilde{r}_1 \sigma_g^2 / 6 > 0$.

Definition 2.5 (Double scaling)**.** Take $\varepsilon \to 0^+$ while holding fixed:

- $\Lambda \equiv \varepsilon \hat{R}_{\mathcal{H}}$, for the factorized kernel;
- $E_0 \equiv \varepsilon E_c \tilde{r}_1$, for the light-cone kernel.

3. Axioms and the field equation

Axiom 1 (Causal accumulation law; C8 of [1])**.**

$$T(x) = \alpha J(x) - (1 - \alpha) \int_{J^-(x)} k(\tau; x, x') \, \mathcal{P}(x, x') \, J(x') \, d^4x'.$$

The kernel k and the transport $\mathcal{P}$ are real. J takes values in the biquaternion carrier $\mathbb{C} \otimes \mathbb{H}$.

Remark 3.1. [2] takes J to be real. That is incompatible with its own phase channel $B \exp(iS/\hbar_{\text{eff}})$, and with any solution proportional to $e^{-iEt/\hbar}$. The reality that [2, Lemma 4.3] actually uses is the reality of $\hat{R}$, and that follows from the reality of k and $\mathcal{P}$.

Axiom 2 (Two channels and channel geometry)**.** These are as in [2, Axioms 2–3]: the channels carry relative orientation $e^{i\theta}$ with $\theta = \alpha\pi/2$.

Axiom 3 (Field equation)**.** On physical configurations $T = 0$, with the memory channel oriented by Definition 2.3:

$$\alpha J(x) = (1 - \alpha) \, e^{-i\sigma\alpha\pi/2} \int_{J^-(x)} k(\tau; x, x') \, \mathcal{P}(x, x') \, J(x') \, d^4x'. \tag{2}$$

Remark 3.2. Axiom 3 replaces Posit 1 of [2]. It assumes no unitarity, no generator, and no identification $\hat{H} = f_{\text{map}}[\hat{T}]$.

The alternative balance law $\partial_t T = 0$ gives the same bounded admissible solutions. If J is admissible then so is T, because $\hat{T}$ is a Fourier multiplier of $\hat{J}$. A constant admissible T has no support at zero frequency, so it vanishes.

4. Unconditional lemmas

Lemma 4.1 (Weyl convergence)**.** *Let $\omega \in \mathbb{R} \setminus \{0\}$. The improper integral $\int_0^\infty s^{\alpha-1} e^{i\omega s}\, ds$ converges if and only if $0 < \alpha < 1$. When it converges, its value is $\Gamma(\alpha)(-i\omega)^{-\alpha}$. Consequently $W^\alpha e^{-iEt/\hbar} = (-iE/\hbar)^{-\alpha} e^{-iEt/\hbar}$ for $0 < \alpha < 1$. At $\alpha = 1$ the right-hand side exists only as the limit $\alpha \to 1^-$.*

Proof. Convergence for $0 < \alpha < 1$. Near $s = 0$ the integrand is integrable. On $[1, \infty)$, $s^{\alpha-1}$ decreases monotonically to 0 and $\int_1^X e^{i\omega s} ds$ is bounded, so Dirichlet's test applies.

Divergence for $\alpha \geq 1$. If $\alpha = 1$, the partial integrals $(e^{i\omega X} - 1)/(i\omega)$ have no limit. If $\alpha > 1$, the amplitude $s^{\alpha-1}$ does not tend to zero, so again there is no limit.

Value. Rotate the contour onto the ray $\arg s = \frac{\pi}{2} \operatorname{sgn} \omega$, and use Jordan's lemma for the arc. □

Lemma 4.2 (Inversion, factorized kernel)**.** *For the factorized kernel, $\mathcal{P}$ is time-translation invariant on the sector, and $\hat{R}_{\mathcal{H}}$ commutes with W^α. On bounded admissible J, (2) is equivalent to*

$$\alpha\, (\sigma i \tau_c)^\alpha\, D^\alpha J = \Lambda J. \tag{3}$$

Proof. On the sector, (2) reads $\alpha J = \varepsilon\, e^{-i\sigma\alpha\pi/2} \tau_c^{-\alpha} \hat{R}_{\mathcal{H}} W^\alpha J$. Apply D^α and multiply by $(\sigma i \tau_c)^\alpha = e^{i\sigma\alpha\pi/2} \tau_c^\alpha$. For the converse, apply W^α. Both maps are injective on admissible functions. □

Lemma 4.3 (Phase cancellation)**.** *Let $g_\alpha(E) \equiv \alpha(\sigma i)^\alpha (-iE/E_c)^\alpha$ be the symbol of the left side of (3). For $E \neq 0$,*

$$g_\alpha(E) = \alpha |E/E_c|^\alpha\, e^{\,i\alpha\pi(\sigma - \operatorname{sgn} E)/2}.$$

For $\alpha \in (0,1) \cup (1,2)$, $g_\alpha(E)$ is real if and only if $\operatorname{sgn} E = \sigma$, and then it is positive and strictly increasing in $|E|$. At $\alpha = 1$, $g_1(E) = \sigma E/E_c$ is real for both signs of E. Finally, setting $f_\alpha(\lambda) \equiv \sigma E_c (\lambda/\alpha)^{1/\alpha}$ for $\lambda \geq 0$ gives $g_\alpha(f_\alpha(\lambda)) = \lambda$.

Proof. The principal arguments are $\arg(\sigma i) = \sigma\pi/2$ and $\arg(-iE) = -\operatorname{sgn}(E)\, \pi/2$. Hence $\arg g_\alpha = \alpha\pi(\sigma - \operatorname{sgn} E)/2 \in \{0, \pm\alpha\pi\}$, and $\pm\alpha\pi \notin \pi\mathbb{Z}$ for $\alpha \notin \{0, 1, 2\}$. The inverse relation follows by direct substitution. □

Lemma 4.4 (Operator facts)**.**

(1) *$\hat{R}_{\mathcal{H}} = r_0 \nabla^2 + r_1$ is self-adjoint on $H^2(\mathbb{R}^3)$: it is unitarily equivalent to multiplication by $r_1 - r_0 |k|^2$. A bounded x-dependent r_1 preserves self-adjointness, by Kato–Rellich [11].*

(2) *$\|\hat{R}_{\mathcal{H}}\| = \infty$, so the spectral inclusion in* [2, Lemma 4.4] *is vacuous.*

(3) *Exchanging the kernel limit with $\Pi_{\mathcal{H}}$ requires strong-resolvent convergence of the projected operators [10]. The bound $\|\Pi_{\mathcal{H}}\| = 1$ alone does not suffice.*

5. Factorized kernel: derived unitarity and the collapse

Write $\mathcal{H}_+ \equiv \operatorname{Ran} \mathbf{1}_{(0,\infty)}(\Lambda)$.

Theorem 5.1 (Bounded solutions are unitary)**.** *Fix $\alpha \in (0,1)$ and σ. Let $J \in C_b(\mathbb{R}; \mathcal{H})$ be admissible and solve (3) in the sense of $\mathcal{H}$-valued tempered distributions. Then:*

(i) $J(t) = e^{-itf_\alpha(\Lambda)/\hbar}\psi$ *for some* $\psi \in \mathcal{H}_+$. *Conversely, every such* J *is a solution.*
(ii) $U_\alpha(t) \equiv e^{-itf_\alpha(\Lambda)/\hbar}\big|_{\mathcal{H}_+}$ *is a strongly continuous one-parameter unitary group with self-adjoint generator* $f_\alpha(\Lambda)$. *In particular* $\|J(t)\| = \|J(0)\|$.
(iii) *Every frequency present satisfies* $\sigma E > 0$, *and* $\operatorname{Ran}\mathbf{1}_{(-\infty,0)}(\Lambda)$ *carries no solution.*

Proof. Step 1: fiber equation. Fourier-transform in t and pass to the spectral representation of Λ. Fiberwise, (3) becomes $\big(g_\alpha(E) - \lambda\big)\hat{J}_\lambda(E) = 0$.

Step 2: zeros. By Lemma 4.3:

- for $\lambda > 0$, the continuous function $g_\alpha - \lambda$ vanishes only at the simple zero $E_\lambda = f_\alpha(\lambda)$, where $\sigma E_\lambda > 0$;
- for $\lambda < 0$, it never vanishes;
- for $\lambda = 0$, its only zero is $E = 0$, which admissibility excludes.

Step 3: support. A tempered distribution annihilated by a continuous function is supported on that function's zero set. So $\hat{J}_\lambda$ is a finite sum of $\delta^{(n)}(E - E_\lambda)$, and $\hat{J}_\lambda = 0$ when $\lambda \le 0$. The terms with $n \ge 1$ produce growth t^n and are excluded by boundedness. Hence $J_\lambda(t) = c_\lambda e^{-iE_\lambda t/\hbar}$, which is (i).

Step 4: converse. By $g_\alpha(f_\alpha(\lambda)) = \lambda$, each such J satisfies the fiber equation. Its spectrum lies in $\sigma E > 0$, so it is admissible.

Step 5: unitarity. Functional calculus makes U_α a strongly continuous unitary group. Stone's theorem [9, 10] identifies its generator as $f_\alpha(\Lambda)$, which is self-adjoint because f_α is real and Borel. Statement (iii) is Step 2. □

Remark 5.2. Theorem 5.1 is the derived replacement for Posit 1 of [2]. The fiberwise argument of Steps 1–3 is stated at the level of rigor of the companion papers. The measurability details are standard but are not written out here.

Theorem 5.3 (Collapse to Schrödinger)**.** *Take the double scaling with* Λ *fixed. For every* $\psi \in \mathcal{H}_+$ *and every* $T > 0$,

$$\sup_{|t|\le T} \big\|U_\alpha(t)\psi - e^{-itH/\hbar}\psi\big\| \xrightarrow[\alpha\to 1^-]{} 0, \qquad H \equiv \sigma E_c \Lambda.$$

If ψ *has compact spectral support in* $(0,\infty)$, *then*

$$\big\|U_\alpha(t)\psi - e^{-itH/\hbar}\psi\big\| \le \frac{|t|}{\hbar}\big\|(f_\alpha - f_1)(\Lambda)\psi\big\|, \qquad (f_\alpha - f_1)(\lambda) = \sigma E_c\, \varepsilon\, \lambda(1 + \ln\lambda) + O(\varepsilon^2). \tag{4}$$

The limiting equation is $i\hbar\,\partial_t J = HJ$. *On the spinless, gauge-neutral sector,*

$$H = -\frac{\hbar^2}{2m}\nabla^2 + V_0, \qquad \frac{\hbar^2}{2m} = -\sigma\varepsilon E_c r_0, \qquad V_0 = \sigma\varepsilon E_c r_1.$$

Positive mass with $r_0 > 0$ *forces* $\sigma = -1$. *Then* $\hbar^2/2m = \varepsilon E_c r_0$, *which is the mass formula of* [2, Thm. 6.1(iii)] *with its units restored, and* $\mathcal{H}_+ = \operatorname{Ran}\mathbf{1}_{(-\infty,0)}(H)$.

Proof. Pointwise limit. Since $f_\alpha \to f_1 = \sigma E_c \lambda$ pointwise on $[0,\infty)$, the integrand $|e^{-itf_\alpha(\lambda)/\hbar} - e^{-itf_1(\lambda)/\hbar}|^2 \le 4$ tends to zero. Dominated convergence in the spectral measure of ψ gives the strong limit. Uniformity for $|t| \le T$ follows from $|e^{ia} - e^{ib}| \le \min(2, |a-b|)$ together with the local uniformity of $f_\alpha \to f_1$.

Rate. The same inequality gives the bound in (4). Expanding $\ln\big((\lambda/\alpha)^{1/\alpha}\big) = (1 + \varepsilon + \dots)(\ln\lambda + \varepsilon + \dots)$ gives the stated difference.

Identification. $f_1(\Lambda) = \sigma E_c \Lambda$, and on the sector Λ acts as multiplication by $\varepsilon(r_1 - r_0|k|^2)$. □

Corollary 5.4 (The skeleton and the edge are two scalings of one limit)**.** (1) *Hold* $\hat{R}_\mathcal{H}$ *fixed, as in* [2, Lemma 4.1]. *Then* $f_\alpha(\varepsilon\hat{R}_\mathcal{H}) \to 0$, *so* $U_\alpha(t) \to \mathbf{1}$ *strongly, and*

$m = \hbar^2/(2\varepsilon E_c r_0) \to \infty$. *The carrier freezes. This is the kernel-blind skeleton of* [2, Thm. 5.1]*, up to its unobservable global phase.*

(2) *Hold* $\Lambda = \varepsilon \hat{R}_{\mathcal{H}}$ *fixed. Then Theorem 5.3 applies: the memory stays live and becomes* H.

The "corner" and the "edge" of [2, Rem. 5.2, Table 7.4] *are therefore one limit taken in two ways.*

Proposition 5.5 (Response to a source)**.** *Let* $\alpha \in (\frac{2}{3}, 1)$*. Let* S *be a bounded source with zero past, and consider* $T = S$*, that is,* (2) *with* S *on the left. Its causal solution is* $\hat{J}(E) = \hat{G}(E)\hat{S}(E)$*, with*

$$\hat{G}(E) = \frac{(\sigma i)^\alpha (-iE/E_c)^\alpha}{g_\alpha(E) - \Lambda},$$

evaluated just above the real axis. On each spectral value $\lambda \neq 0$*,* $\hat{G}$ *is analytic in* $\operatorname{Im} E > 0$*; its other singularities are zeros of* $g_\alpha - \lambda$ *and the branch cut on the negative imaginary axis.*

(1) *For* $\lambda > 0$ *the zero is the real point* $E_\lambda = f_\alpha(\lambda)$*. After the source has acted, the response is the stationary solution of Theorem 5.1, with amplitude*

$$\Big| \frac{(\lambda/\alpha)\, \hat{S}(E_\lambda)}{g'_\alpha(E_\lambda)} \Big|,$$

plus a tail from the branch cut that decays as a power of t*.*

(2) *For* $\lambda < 0$ *the zero is at* $\operatorname{Im} E = -\pi\varepsilon|E|\big(1 + O(\varepsilon)\big) < 0$*. Such a mode exists only as a causal transient that decays at this rate. It has no bounded solution on the whole time line, consistent with Theorem 5.1(iii).*

Proof. Response function. Fourier-transform $\alpha J - \varepsilon e^{-i\sigma\alpha\pi/2} \tau_c^{-\alpha} \hat{R}_{\mathcal{H}} W^\alpha J = S$ and multiply by $(\sigma i)^\alpha (-iE/E_c)^\alpha$. This gives $\hat{G}$.

Analyticity. The factor $(-iE)^\alpha$, principal branch, is analytic away from $-iE \in (-\infty, 0]$, which is the negative imaginary E-axis. This analyticity in the upper half-plane is the causality of W^α.

Location of the zeros. Put $w = -iE/E_c$, so that $\operatorname{Im} E = E_c \operatorname{Re} w$, and $g_\alpha = \lambda$ becomes $w^\alpha = (\lambda/\alpha)(\sigma i)^{-\alpha}$.

- For $\lambda > 0$: $w = -i\sigma(\lambda/\alpha)^{1/\alpha}$, so $\operatorname{Re} w = 0$ and the zero is real.
- For $\lambda < 0$: the principal root has $\arg w = \pm(\pi/\alpha - \pi/2)$. A principal root exists iff $\pi - \alpha\pi/2 < \alpha\pi$, that is, $\alpha > \frac{2}{3}$. Then $\operatorname{Re} w = |w| \sin(\pi/\alpha)$, and for $\alpha = 1 - \varepsilon$ this is $-|w| \sin(\pi\varepsilon)\,(1 + O(\varepsilon))$.

Late-time response. For t after the source, close the inversion contour in the lower half-plane. The residue at the real zero gives (1). The cut gives the power-law tail [6, 5]. □

Remark 5.6. A Caputo initial-value problem with $J(0) = J_0$ corresponds to the frozen past $J \equiv J_0$ for $t < 0$. Its stationary amplitude is $\alpha^{-1}|J_0|$, not $|J_0|$. That factor belongs to the frozen past. The frozen past is not admissible for Axiom 3, so the factor is not a prediction of the field equation.

Theorem 5.7 (The limit is one-sided)**.** (1) Convergence. W^α *converges on oscillating carriers only for* $\alpha < 1$ *(Lemma 4.1). The memory term of* (2) *at* $\alpha = 1$ *exists only as the limit from below.*

(2) Sign. *With* σ *fixed, the memory-generated coefficients* $\hbar^2/2m = -\sigma\varepsilon E_c r_0$ *(factorized) and* $E_0 = \varepsilon E_c \tilde{r}_1$ *(light cone) are odd in* ε*. Positivity can hold on one side of* $\alpha = 1$ *only. For the light-cone kernel with* $\tilde{r}_1 > 0$*, there are no stationary solutions at all for* $\alpha > 1$ *(§6).*

(3) Runaway or preacceleration. *For $\alpha \in (1,2)$, take the source problem with zero past. There the Riemann–Liouville integral is finite, and $\hat{G}$ is defined by the continued symbol. On* $\operatorname{Ran} \mathbf{1}_{(-\infty,0)}(\Lambda)$*, $g_\alpha - \lambda$ has a zero with* $\operatorname{Im} E \approx +\pi|\varepsilon||E| > 0$*, in the upper half-plane. Then:*
- *the causal response, obtained by inverting above every singularity, grows at that rate;*
- *the bounded response, obtained by inverting on the real axis, is nonzero before the source acts.*

This is the Abraham–Lorentz–Dirac dichotomy of the third-derivative panel of [1]*, runaway or preacceleration, appearing on the wrong side of $\alpha = 1$.*

Proof. Part (1) is Lemma 4.1.

For part (2), the light-cone statement follows from Lemma 6.3 below. There the symbol is real and positive on the stationary sector for every $\alpha \in (0,2)$. So the plane-wave equation $\alpha = \varepsilon\tilde{r}_1 \times$ (positive) has no solution when $\varepsilon\tilde{r}_1 < 0$.

For part (3), the argument in the proof of Proposition 5.5 gives $\operatorname{Re} w = |w| \sin(\pi/\alpha)$, which is positive for $\alpha \in (1,2)$. A zero of the response function in the upper half-plane rules out a response that is both causal and bounded. □

6. Memory inside the past light cone

6.1. **Markovian evolution, and why the cone still matters.** The limit taken here is Markovian, so one may ask why the past light cone matters at all. There are two separate notions.

(1) *Markovian flow.* The bounded solutions form a one-parameter group, so the state at one time determines the state at every other time. By Theorem 5.1 this already holds for every $\alpha < 1$, and it holds in the limit. Memory survives only in the dispersion relation and in the response to sources (Proposition 5.5), not in the flow.

(2) *δ-kernel.* The lag weight tends to $\delta(\tau)$.

The limit $\alpha \to 1^-$ is Markovian in sense (1) but not in sense (2). At $\alpha = 1$ the Weyl kernel is the flat accumulator $\Theta(\tau)$, which weights the entire past equally. Its inverse is ∂_t, exactly as $X(t) - X(t_0) = \int_{t_0}^{t} \dot{X}$. The generator is local in time because accumulation and differentiation are inverse to each other, not because the kernel is narrow.

The light cone persists because Axiom 1 integrates over $J^-(x)$. Any spatial structure, and in particular the kinetic term, has to come from the memory's spatial reach. Inside the cone that reach is at most c times the lag.

Lemma 6.1 (No kinetic term from a causal δ-kernel)**.** *Let the lag weight $w \geq 0$ be normalized and compactly supported, and let each g_τ be isotropic, normalized, and supported in $|y| \leq c\tau$. Then*

$$\int w(\tau)\, \hat{g}_\tau(k)\, d\tau = 1 - \frac{k^2\rho^2}{6} + O(k^4), \qquad \rho^2 \equiv \int w(\tau)\, \langle |y|^2 \rangle_{g_\tau}\, d\tau \leq c^2 \int \tau^2 w(\tau)\, d\tau.$$

In particular, as $w \to \delta$ the coefficient of ∇^2 tends to zero.

Proof. For an isotropic profile, $\hat{g}_\tau(k) = 1 - \frac{1}{2}\langle (k\cdot y)^2 \rangle + O(k^4)$, and $\langle (k\cdot y)^2 \rangle = k^2 \langle |y|^2 \rangle / 3$. Then use $\langle |y|^2 \rangle_{g_\tau} \leq c^2\tau^2$. □

The δ-kernel limit of [2] with $r_0 \neq 0$ therefore assumes instantaneous spatial reach, which is outside the past light cone. The kinetic term, and with it inertia, needs memory at nonzero lags. The accumulator of the $\alpha \to 1^-$ limit supplies exactly that while keeping the flow Markovian. The cone then fixes the mass (Proposition 6.5). The factorized kernel of §5, whose spread does not shrink with the lag, violates the cone at short lags. The physical kernel class is the light-cone-supported one.

6.2. **The light-cone dispersion.** On the spinless sector, (2) on a plane wave $e^{-iEt/\hbar+ik\cdot x}$ reads

$$\alpha = \varepsilon\tilde{r}_1\, e^{-i\sigma\alpha\pi/2}\,\tau_c^{-\alpha}\, m_\alpha(E,k), \qquad m_\alpha(E,k) \equiv \frac{1}{\Gamma(\alpha)}\int_0^\infty \tau^{\alpha-1} e^{iE\tau/\hbar}\,\hat{g}_\tau(k)\,d\tau. \tag{5}$$

Proposition 6.2 (Thin-shell symbol)**.** *For a thin shell, $\hat{g}_\tau(k) = \sin(kv\tau)/(kv\tau)$. With $s = -iE/\hbar$ and $0 < \alpha < 1$,*

$$m_\alpha(E,k) = \frac{(s-ikv)^{1-\alpha} - (s+ikv)^{1-\alpha}}{2ikv\,(\alpha-1)},$$

and as $\alpha \to 1^-$ this tends to $m_1 = \dfrac{i}{kv}\operatorname{artanh}\Big(\dfrac{\hbar kv}{E}\Big)$ for $E > \hbar kv$.

Proof. Write $\sin(kv\tau) = (e^{ikv\tau} - e^{-ikv\tau})/(2i)$. Use $\int_0^\infty \tau^{\beta-1}e^{-p\tau}d\tau = \Gamma(\beta)p^{-\beta}$, continued to $\beta = \alpha - 1$. The combined integrand $\tau^{\alpha-2}\sin(kv\tau)$ is integrable at 0 and converges at ∞ by Dirichlet's test. Then use $\Gamma(\alpha-1)/\Gamma(\alpha) = (\alpha-1)^{-1}$. The limit follows from $(a^\varepsilon - b^\varepsilon)/\varepsilon \to \ln(a/b)$. □

Lemma 6.3 (Phase cancellation in the cone)**.** *Let the profile be self-similar and supported in $|y| \le c\tau$, and let $E > \hbar kc$. For every $\alpha \in (0,2)\setminus\{1\}$:*

- *with $\sigma = +1$, the quantity $e^{-i\sigma\alpha\pi/2}m_\alpha(E,k)$ is real and positive;*
- *with $\sigma = -1$, it is not real.*

The same holds as $\alpha \to 1$. Stationary solutions on $E > \hbar kc$ therefore exist only for $\sigma = +1$, and they are unitary. The case $\sigma = -1$ is the mirror sector $E < -\hbar kc$.

Proof. One shell. Take a shell of speed $u \le c$. Then $s - iku = -i(E/\hbar + ku)$ and $s + iku = -i(E/\hbar - ku)$, and both are $-i$ times a positive number. So

$$m_\alpha = e^{-i\pi(1-\alpha)/2}\,\frac{(E/\hbar+ku)^{1-\alpha} - (E/\hbar-ku)^{1-\alpha}}{2iku(\alpha-1)} = e^{i\pi\alpha/2}P,$$

where $P \equiv -\big[(E/\hbar+ku)^{1-\alpha} - (E/\hbar-ku)^{1-\alpha}\big]/\big(2ku(\alpha-1)\big) > 0$ for both $\alpha < 1$ and $\alpha > 1$, because the bracket and $\alpha - 1$ always have opposite signs. Multiplying by $e^{-i\sigma\alpha\pi/2}$ gives $e^{i\pi\alpha(1-\sigma)/2}P$.

General profile. A self-similar profile is a nonnegative superposition of shells over $u \in [0,c]$. Each shell contributes a positive real number when $\sigma = +1$, so the sum is positive real. □

Theorem 6.4 (Light-cone dispersion at $\alpha \to 1^-$)**.** *Take the thin shell with $\sigma = +1$, and the double scaling with E_0 fixed. As $\alpha \to 1^-$, the stationary energies converge to the unique root with $E > \hbar kv$ of $\operatorname{artanh}(\hbar kv/E) = \hbar kv/E_0$, namely*

$$E(k) = \hbar kv\,\coth\Big(\frac{\hbar kv}{E_0}\Big) = E_0 + \frac{\hbar^2k^2v^2}{3E_0} - \frac{\hbar^4k^4v^4}{45E_0^3} + \dots \tag{6}$$

(1) Schrödinger limit. *For $\hbar kv \ll E_0$, $E = E_0 + \hbar^2k^2/2m$ with $m = 3E_0/(2v^2)$. The relative correction to the kinetic term is $-y^2/15$, where $y \equiv \hbar kv/E_0$.*

(2) Causality. *The group velocity $v_g = v\,(\coth y - y\,\operatorname{csch}^2 y)$ satisfies $0 < v_g < v \le c$ for all k.*

Proof. The root. At $\alpha = 1$, Proposition 6.2 gives $e^{-i\pi/2}m_1 = \operatorname{artanh}(\hbar kv/E)/(kv)$. Also $\varepsilon\tilde{r}_1/\tau_c = E_0/\hbar$. So (5) becomes $\operatorname{artanh}(\hbar kv/E) = \hbar kv/E_0$. Since artanh is a bijection from $(0,1)$ to $(0,\infty)$, the root is unique, and it equals $\hbar kv/\tanh(\hbar kv/E_0) > \hbar kv$.

Convergence in α. The left side of (5) converges locally uniformly in E as $\alpha \to 1^-$. Its E-derivative at the root is nonzero. By the implicit function theorem the roots converge.

Expansion. The series follows from $x\coth x = 1 + x^2/3 - x^4/45 + \dots$.

Group velocity. $\coth y - y\,\mathrm{csch}^2 y = (\sinh y \cosh y - y)/\sinh^2 y$. The numerator is positive because $\sinh(2y) > 2y$. The quotient is less than 1 because it is equivalent to $\sinh y\, e^{-y} < y$, i.e. $1 - e^{-2y} < 2y$. □

Proposition 6.5 (General light-cone profile)**.** *For a self-similar profile supported in the light cone, as* $\alpha \to 1^-$,

$$E(k) = E_0 + \frac{\hbar^2 k^2}{2m} + O(k^4), \qquad m = \frac{3E_0}{2v_{\mathrm{eff}}^2}.$$

Proof. Superposing shells, the equation becomes $1 = (E_0/E)\int w(u)\, A(\hbar k u/E)\, du$ with $A(x) = \operatorname{artanh}(x)/x = 1 + x^2/3 + O(x^4)$. This gives $E = E_0 + \hbar^2 k^2 v_{\mathrm{eff}}^2/(3E_0) + O(k^4)$. □

Corollary 6.6 (Zero-momentum energy of causal memory)**.** $E_0 = \frac{2}{3}\, m\, v_{\mathrm{eff}}^2 \le \frac{2}{3}\, mc^2$*, with equality only when all the memory sits on the light cone. The zero-momentum energy generated by light-cone-supported memory is therefore not the rest energy* mc^2*. Moreover, the mass is not a free parameter. It is fixed by the zeroth moment and by the fraction of the light cone that the memory fills, which sharpens* [2, Thm. 6.1(iii)].

Proposition 6.7 (Approach from below, thin shell)**.** *For* $\alpha < 1$ *and* $\sigma = +1$*:*

1. $E(0) = E_c\big(E_0/(\alpha E_c)\big)^{1/\alpha}$*;*
2. *the effective mass is* $m_\alpha = 3E(0)/\big((1+\alpha)v^2\big)$*;*
3. *a stationary root exists whenever* $\hbar k v < \frac{1}{2}E_c\big(E_0/(\alpha\varepsilon E_c)\big)^{1/\alpha}$*, a cutoff that recedes to infinity as* $\varepsilon \to 0$*.*

Proof. Small k*.* Expand $\hat g_\tau = 1 - k^2 v^2 \tau^2/6 + O(k^4\tau^4)$. Then $\Gamma(\alpha)^{-1}\int \tau^{\alpha+1} e^{-s\tau} d\tau = \alpha(\alpha+1)s^{-\alpha-2}$ and $s^{-2} = -\hbar^2/E^2$. With Lemma 6.3,

$$(E/E_c)^\alpha = \frac{E_0}{\alpha E_c}\Big[1 + \frac{\alpha(\alpha+1)\hbar^2 k^2 v^2}{6E^2}\Big] + O(k^4).$$

Setting $k = 0$ gives (1), and linearizing in k^2 gives (2).

Existence. As $E \to \hbar k v^+$, the right side of (5) tends to $\tilde r_1 (2kv\tau_c)^{-\alpha}$. As $E \to \infty$ it tends to 0. The intermediate value theorem gives a root whenever $\alpha < \tilde r_1 (2kv\tau_c)^{-\alpha}$, which is (3). □

Remark 6.8. The dispersion (6) is defined in the frame of the foliation, and it is not Lorentz invariant. No relativistic claim is made. What it does supply is a causal completion of the Schrödinger dispersion. It is valid for particle speeds $u \ll v_{\mathrm{eff}}$, and its corrections are of order $(u/v_{\mathrm{eff}})^2$.

7. Numerical verification

All statements above were run with the script `scripts/cal_schrodinger_collapse.py` in the repository https://github.com/JewJitsu11B/Liorhybrid. It uses NumPy only and runs in about one minute. Units are $\hbar = \tau_c = 1$, so $E_c = 1$. Methods are described in Appendix B.

Factorized kernel, with $\sigma = -1$, $m = 1$, and cap $E_{\mathrm{cap}} = 4$. The test state is a Gaussian packet with $k_0 = 1$ and width 2; its weight above the cap is 10^{-13}.

Source-driven runs. The integral equation $T = S$ is time-stepped directly, with Weyl integration over a zero past and the full history kept. The source is a Gaussian pulse at $t_0 = 10$ of width 2, tuned to the mode. No initial-value data enter.

Light-cone kernel, with a thin shell, $v = 1$, $E_0 = 1$ and $\sigma = +1$. On $E > \hbar k v$, the measured ratio $|\mathrm{Im}|/|\cdot|$ of the symbol is at most 7×10^{-15} for $\alpha \in \{0.7, 0.9, 0.99, 1\}$, as Lemma 6.3 requires. With $\tilde r_1 > 0$ and $\alpha \in \{1.001, 1.01\}$, no stationary root exists.

ε	$\max\lvert E_\alpha - E_1\rvert$	$\varepsilon \max\lambda\lvert 1+\ln\lambda\rvert$	ratio	phase-aligned error, $t \le 40$	norm drift
10^{-1}	1.246	0.954	1.305	1.41	3×10^{-16}
10^{-2}	9.78×10^{-2}	9.55×10^{-2}	1.025	3.30×10^{-1}	3×10^{-16}
10^{-3}	9.57×10^{-3}	9.55×10^{-3}	1.002	3.27×10^{-2}	3×10^{-16}
3×10^{-4}	2.87×10^{-3}	2.86×10^{-3}	1.001	9.80×10^{-3}	3×10^{-16}

TABLE 1. Bounded solutions (Theorems 5.1–5.3). The energy error matches the first-order term of (4), and the norm is conserved to machine precision. The raw packet error is dominated by a global phase drift from the constant term. With that phase removed, the error is linear in ε, at about $33\,\varepsilon$ over $t \le 40$.

α	h	mean $\lvert J\rvert$, $t\in[80,120]$	residue prediction	ratio
0.900	0.02	6.95236	6.95787	0.99921
0.900	0.01	6.95674	6.95787	0.99984
0.990	0.01	5.16659	5.16724	0.99987
0.999	0.01	5.02771	5.02833	0.99988
1.000	0.01	5.01264	5.01326	0.99988
1.010	0.01	4.86571	4.86630	0.99988

TABLE 2. Below the cap ($\Lambda = +1$), Proposition 5.5(1). After the pulse the mode oscillates forever at the amplitude of the residue at the real zero. This is the stationary solution of Theorem 5.1, excited causally. The ratio converges as h is halved.

α	zero of $g_\alpha - \Lambda$	$-\operatorname{Im} E$	$\pi\varepsilon\lvert E\rvert$	measured decay rate
0.970	$1.027032 - 0.100104\,i$	$+1.0010\times10^{-1}$	$+9.73\times10^{-2}$	$+1.0007\times10^{-1}$
0.990	$1.009695 - 0.032052\,i$	$+3.2052\times10^{-2}$	$+3.17\times10^{-2}$	$+3.2049\times10^{-2}$
0.999	$1.000997 - 0.003148\,i$	$+3.1479\times10^{-3}$	$+3.14\times10^{-3}$	$+3.1478\times10^{-3}$
1.000	1.000000	0	0	$+2.5\times10^{-7}$
1.001	$0.998997 + 0.003135\,i$	-3.1353×10^{-3}	-3.14×10^{-3}	-3.1348×10^{-3}
1.010	$0.989718 + 0.030795\,i$	-3.0795×10^{-2}	-3.11×10^{-2}	-3.0792×10^{-2}
1.030	$0.967645 + 0.088790\,i$	-8.8790×10^{-2}	-9.16×10^{-2}	-8.8782×10^{-2}

TABLE 3. Above the cap ($\Lambda = -1$), Proposition 5.5(2) and Theorem 5.7(3). From below the zero lies in $\operatorname{Im} E < 0$, giving causal decay. At $\alpha = 1$ it lies on the real axis. From above it lies in $\operatorname{Im} E > 0$, and the causal response grows. The measured rate is $(\ln\lvert J(40)\rvert - \ln\lvert J(80)\rvert)/40$.

α	$\lvert J(t_0-10)\rvert$	$\lvert J(t_0-20)\rvert$	$\lvert J(t_0+40)\rvert$
0.99	4.0×10^{-6}	7.5×10^{-11}	1.44
1.01	3.58	2.63	4.2×10^{-6}

TABLE 4. Bounded responses, from inversion on the real axis, for $\Lambda = -1$. From below the bounded response is causal: what appears before the pulse is only the Gaussian's own tail. From above, keeping the response bounded makes it precede its source. That is preacceleration, Theorem 5.7(3).

k	root	$kv\coth(kv/E_0)$	$E_0 + k^2/2m$	rel. kinetic dev.	K/E_0	v_g/v
0.01	1.000033	1.000033	1.000033	-6.7×10^{-6}	0.000	0.007
0.10	1.003331	1.003331	1.003333	-6.7×10^{-4}	0.003	0.067
0.30	1.029822	1.029822	1.030000	-6.0×10^{-3}	0.030	0.198
1.00	1.313035	1.313035	1.333333	-6.1×10^{-2}	0.333	0.589
3.00	3.014909	3.014909	4.000000	-0.33	3.000	0.975

TABLE 5. Theorem 6.4. The numerical root agrees with (6). The deviation from Schrödinger matches $-y^2/15$, and the group velocity stays below v.

ε	$E(0)$	$(E_0/\alpha)^{1/\alpha}$	m (fit)	$3E(0)/((1+\alpha)v^2)$	$\max_{k\le 1}\lvert E_\alpha - E_1\rvert$	cutoff kv
10^{-1}	1.124195	1.124195	1.77507	1.77504	1.24×10^{-1}	7.3
10^{-2}	1.010204	1.010204	1.52296	1.52292	1.02×10^{-2}	53
10^{-3}	1.001002	1.001002	1.50229	1.50225	1.00×10^{-3}	504
10^{-4}	1.000100	1.000100	1.50026	1.50023	1.00×10^{-4}	5010

TABLE 6. Proposition 6.7. The approach from below is linear in ε, and $m \to 3E_0/(2v^2) = 1.5$.

8. Remarks on scope

Remark 8.1 (Status ledger). *Inputs:*

- the field equation (Axiom 3);
- the kernel class (Definition 2.2);
- the double scaling (Definition 2.5).

Derived from these: every lemma, theorem, proposition and corollary of §§4–6.
Not claimed:

- a value for E_0;
- Lorentz invariance of (6);
- gauge coupling, where transports with $U(1)$ holonomy make $\hat{R}_{\mathcal{H}}$ Hermitian rather than real;
- position-dependent potentials beyond bounded $r_1(x)$;
- any initial-value (Caputo) statement, since a frozen past is not admissible.

Remark 8.2 (What changes relative to [2]).

- Posit 1, and Theorem 6.1(i) built on it, are replaced by Axiom 3 together with Theorems 5.1 and 5.3.
- Axiom 1 should require k and $\mathcal{P}$ to be real, not J.
- In Lemma 4.3, real plus symmetric gives a symmetric operator; self-adjointness is Lemma 4.4(1).
- In Lemma 4.4, $\|\hat{R}\| = \infty$, and exchanging the limit with the projection needs strong-resolvent convergence.
- The mass formula of Theorem 6.1(iii) holds in the double scaling. At fixed r_0 the mass diverges, which is Corollary 5.4(1).
- The constant shift in Eq. (11) is inert only at $\alpha = 1$. For $\alpha < 1$ it sets the stationary sector, by Theorem 5.1(iii).
- The 1^- edge of Table 7.4 is one-sided by Theorem 5.7, not by convention.
- The δ-kernel limit with $r_0 \neq 0$ needs instantaneous spatial reach (Lemma 6.1). The Markovian limit is reached through the accumulator, not through a δ-kernel.

Remark 8.3 (Dynamics is memory). Corollary 5.4 makes [2, Rem. 7.3] precise. When the memory weight vanishes with the memory operator held fixed, the carrier freezes. When the memory weight vanishes with its product with the memory operator held fixed, the memory becomes the Hamiltonian. The Schrödinger generator is the image of the accumulated past under the double scaling.

Appendix A. Notation

α, $\varepsilon = 1 - \alpha$	fractional order and memory weight
σ, $\theta = \alpha\pi/2$	channel orientation and channel angle
W^α, D^α	causal Weyl integral and its inverse (Definition 2.1)
τ_c, $E_c = \hbar/\tau_c$	memory time unit and its energy
g_τ, v, v_{eff}	spatial profile at lag τ, shell speed, effective speed
$\hat{R}_{\mathcal{H}} = r_0\nabla^2 + r_1$	projected memory operator (factorized kernel)
$\Lambda = \varepsilon\hat{R}_{\mathcal{H}}$, $E_0 = \varepsilon E_c \tilde{r}_1$	double-scaled memory operator and offset
g_α, f_α	symbol of the left side of (3), and stationary energy
$\mathcal{H}_+$	$\operatorname{Ran} \mathbf{1}_{(0,\infty)}(\Lambda)$, the stationary sector
$m_\alpha(E, k)$	memory symbol, light-cone kernel (5)

Appendix B. Numerical methods

Time stepping. The Weyl integral over a zero past is discretized by BDF2 convolution quadrature [7], with weights q_j generated by $\big((1-\zeta)+(1-\zeta)^2/2\big)^{-\alpha}$. The full history is kept. Each step solves

$$(\alpha - \kappa h^\alpha q_0)\, J_n = S_n + \kappa h^\alpha \sum_{j\geq 1} q_j J_{n-j}, \qquad \kappa = \Lambda e^{-i\sigma\alpha\pi/2}.$$

This is the integral field equation itself. No derivative and no initial-value data are used.

Zeros and residues. Zeros of $g_\alpha - \Lambda$ are found by Newton iteration. Residue amplitudes use $\hat{S}(E) = \sqrt{2\pi}\, w\, e^{-w^2(E-\Omega)^2/2} e^{i(E-\Omega)t_0}$. The bounded responses of the last table come from direct inversion on the real axis, on a grid of 1.6×10^5 points over $|E| \leq 40$.

Packets. Packets use a periodic grid of 2048 points on a box of length 400, with a unitary FFT. Modes with amplitude below 10^{-10} of the maximum are dropped.

Light cone. Stationary roots on $E > \hbar k v$ are found by bisection on the real part of (5), after checking that the imaginary part vanishes (Lemma 6.3).

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